The Geometry of Certified Policy Improvement Under Function Approximation

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Abstract

Policy optimization algorithms — including TRPO, PPO, and Conservative Policy Iteration — derive their theoretical guarantees from the Performance Difference Lemma (PDL), which requires exact Q-function values unavailable in practice. When the critic f approximates Q^{μ_k} rather than equaling it, a computable quantity we call the *certification gap* $\hat{\Delta}_k := J(\mu_k) - \nu^\top f_{k+1}$ governs whether any improvement guarantee holds. We develop a complete theory centered on this object.

(I) Generalized PDL. For any critic f_{k+1}, any policy update, and any MDP, the exact identity $J(\mu_{k+1}) - J(\mu_k) = A - \hat{\Delta}_k$ holds, where A is what CRPI bounds (improvement over the certified floor) and $\hat{\Delta}_k$ is the exact gap between them. Both terms are computable from rollout data.

(II) Phase Transition. The CRPI improvement bound undergoes a hard phase transition at the improvement ratio $\tilde{\rho}_k := 2(1-\gamma)A_k/(C_k\epsilon_u) = 1$. Below this threshold, no step size certifies improvement. The certified efficiency above the threshold is exactly $\eta_k = (1 - 1/\tilde{\rho}_k)^2$. We correct a factor-of-γ error in prior work: the correct failure region is $\{\tilde{\rho}_k \leq 1\}$, strictly larger than the previously reported $\{\rho_k \leq 1\}$.

(III) Relative Criterion. Fixed absolute critic tolerances are structurally insufficient: as the policy improves and $A_k \to 0$, any fixed ϵ_u eventually dominates, driving $\tilde{\rho}_k$ below 1 and freezing improvement. The correct stopping criterion scales with the current advantage: $\epsilon_u \leq c \cdot A_k$. Under this criterion, monotone convergence to the optimal policy is guaranteed.

(IV) TV Geometry. Under function approximation, the dominant error in the improvement bound is first-order in step size, controlled by total variation distance. TRPO's KL trust region constrains a second-order quantity that is irrelevant at the scale where FA error dominates. The TV-natural gradient $M(\theta)^{-1}g^f$ strictly dominates TRPO's natural gradient $F(\theta)^{-1}g^f$ in certified improvement ratio, with equality only at the uniform policy.

We introduce the **Certified TD Critic (CTDC)**, which converges to $\hat{\Delta}_k^* = 0$ at its fixed point (Condition C), and the $\tilde{\rho}$-adaptive Algorithm 1, which provides certified per-step improvement throughout training. All four contributions are verified experimentally to machine precision in the tabular setting.

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1 Introduction

The Performance Difference Lemma (PDL) of Kakade and Langford [2002] states:

$$J(\pi') - J(\pi) = \frac{1}{1-\gamma}\,\mathbb{E}_{(s,a)\sim d^{\pi'}}\left[A^{\pi}(s,a)\right], \tag{1}$$

where A^{π} is the advantage function of π and $d^{\pi'}$ is the discounted occupancy of π'. This identity underlies Conservative Policy Iteration [CPI; Kakade and Langford, 2002], TRPO [Schulman et al., 2015], PPO [Schulman et al., 2017], and Safe Policy Iteration [Metelli et al., 2021]. Each algorithm constructs a surrogate objective from this identity, bounds the approximation error introduced by restricting the policy change, and derives per-step improvement guarantees. The central assumption is exact knowledge of Q^{π}.

This assumption is never satisfied in practice. The critic f approximates Q^{μ_k} but does not equal it. The consequences are empirically severe — PPO and TRPO exhibit instability and performance collapse across a wide range of settings — but the *exact manner* in which critic approximation error enters the improvement guarantee has remained incompletely characterized.

The central object: the certification gap. We show that critic approximation error is captured exactly by a single computable quantity:

$$\hat{\Delta}_k := J(\mu_k) - \nu^{\top} f_{k+1}. \tag{2}$$

This is the signed bias of the critic under the initial distribution ν. It is positive when the critic underestimates $J(\mu_k)$, negative when it overestimates, and zero only when the critic is exactly correct. Crucially, $\hat{\Delta}_k$ is *computable from rollout data* without knowing Q^{μ_k}: it equals the average Bellman residual of f_{k+1} under μ_k, weighted by the discounted occupancy.

Our main identity (Theorem 7) states that for any critic, any policy update, and any MDP:

$$J(\mu_{k+1}) - J(\mu_k) = A - \hat{\Delta}_k, \tag{3}$$

where A is what the CRPI bound controls (improvement over the critic floor) and $\hat{\Delta}_k$ is the exact accounting for how much this overstates actual improvement. The classical PDL is the special case $\hat{\Delta}_k = 0$.

Four questions, one object. Once $\hat{\Delta}_k$ is identified as the central object, four questions become natural:

Q1. What is the exact form of policy improvement under an approximate critic? *Answer: Theorem 7.*

Q2. When does the CRPI bound certify improvement, and how efficiently? *Answer: Theorems 20 and 22.*

Q3. What critic training objective achieves $\hat{\Delta}_k \to 0$? *Answer: CTDC (Definition 38, Theorem 14).*

Q4. What is the correct optimization geometry for gradient updates under FA? *Answer: Theorem 36.*

Summary of contributions.

1. **Generalized PDL** (Section 3): The exact identity (3) for any critic. $\hat{\Delta}_k$ is both the certification gap and a rollout-estimable quantity. Verified to machine precision across 500 random instances (Figure 3).

2. **Phase transition and efficiency formula** (Section 4): Hard phase transition at $\tilde{\rho}_k = 1$. Efficiency formula $\eta_k = (1 - 1/\tilde{\rho}_k)^2$. Correction of a factor-of-γ error in prior work. Verified exactly (Figure 1).

3. **Relative critic criterion** (Section 4.4): Fixed absolute tolerances structurally fail as $A_k \to 0$. The correct criterion $\epsilon_u \leq c \cdot A_k$ guarantees monotone convergence (Theorem 28). Demonstrated with a visible performance gap when the criterion is violated (Figure 7).

4. **TV geometry for gradient updates** (Section 5): FA introduces first-order errors in step size; TRPO constrains second-order errors. The TV-natural gradient strictly dominates TRPO's direction. Verified analytically (Figure 6).

Relationship to prior work. The Reliable Policy Iteration (RPI) framework of Eshwar et al. [2025b] introduces the RPI invariant ($T^{\mu_k} f \geq f$ coordinatewise), proves $Q^{\mu_k} \geq f$ under this invariant, and derives per-step improvement bounds via CRPI. Our contributions stand in the following relationship:

- Theorem 7 makes explicit a structural identity implicit in their Lemma 3.4, and gives the exact two-term decomposition of actual policy improvement.
- Theorems 20–**??** analyze the CRPI bound's structure to derive the phase transition, efficiency formula, and relative criterion, and correct a threshold error in their Remark 5.2.
- Theorem 36 shows that TV geometry dominates KL — a consequence invisible at the level of the CRPI bound alone.
- CTDC (Definition 38) replaces the RPI push-up loss, which converges to a negative $\hat{\Delta}^*$ violating the certification requirement, with on-policy TD augmented by a one-sided Bellman ceiling.

Organization. Section 2 establishes notation. Section 3 proves the Generalized PDL. Section 4 analyzes the phase transition, efficiency loss, and relative criterion. Section 5 develops the TV geometry for gradient policy updates. Section 6 introduces CTDC. Section 7 presents the experiments. Section 8 states precisely what is exact and what is approximate. Section 9 surveys related work. Section 10 concludes.

2 Notation and Preliminaries

We work in a finite MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with $|\mathcal{S}| = S$, $|\mathcal{A}| = A$, transition kernel $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, reward $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$, and discount $\gamma \in [0, 1)$.

Flat indexing. All Q-functions are vectors in $\mathbb{R}^{SA}$ indexed by $i = s \cdot A + a$.

Bellman operator. For policy $\mu : \mathcal{S} \to \Delta(\mathcal{A})$:

$$(T^\mu f)(s,a) = r(s,a) + \gamma \sum_{s'} P(s'|s,a) \sum_{a'} \mu(a'|s') f(s',a').$$

The $SA \times SA$ stochastic matrix P^μ has entries $P^\mu[(s,a),(s',a')] = P(s'|s,a)\mu(a'|s')$, giving $T^\mu f = r + \gamma P^\mu f$.

Q-function and performance. $Q^\mu = (I - \gamma P^\mu)^{-1} r \in \mathbb{R}^{SA}$. Invertibility holds since the eigenvalues of P^μ satisfy $|\lambda| \leq 1$, making $I - \gamma P^\mu$ invertible. For $\nu \in \Delta(\mathcal{S} \times \mathcal{A})$: $J(\mu) = \nu^\top Q^\mu$.

Discounted occupancy.

$$d^\mu_\nu := (1-\gamma)\, \nu^\top (I - \gamma P^\mu)^{-1} \in \mathbb{R}^{SA}.$$

We write d^μ when ν is clear.

Structural identity. From the definition of d^μ:

$$(d^\mu)^\top (I - \gamma P^\mu) = (1-\gamma)\, \nu^\top. \tag{$\star$}$$

This identity is used throughout.

State advantage function.

$$a^{\mu'}_\mu(f)(s,a) = \sum_{s'} P(s'|s,a) \sum_{a'} [\mu'(a'|s') - \mu(a'|s')] f(s',a') = [(P^{\mu'} - P^\mu) f](s,a).$$

Certified advantage. $A^{\mu'}_\mu(f) = (d^\mu)^\top a^{\mu'}_\mu(f)$.

Weighted TV distance.

$$\|\mu' - \mu\|_{1,\delta^\mu} = \sum_s \delta^\mu(s)\, \|\mu'(\cdot|s) - \mu(\cdot|s)\|_1,$$

where $\delta^\mu(s') = \sum_{s,a} d^\mu(s,a) P(s'|s,a)$ is the one-step reachable state distribution.

Span. For $x \in \mathbb{R}^{SA}$: $^(x) = \max_{s,a} x(s,a) - \min_{s,a} x(s,a)$.

RPI invariant. Critic f satisfies the *RPI invariant* with respect to μ if $T^\mu f \geq f$ coordinatewise. By Eshwar et al. [2025b, Claim 4.1], this implies $Q^\mu \geq f$ coordinatewise.

Uniform Bellman residual. For critic f satisfying the RPI invariant w.r.t. μ:

$$\epsilon_u(f,\mu) = \|T^\mu f - f\|_\infty = \max_{s,a} [(T^\mu f)(s,a) - f(s,a)] \geq 0.$$

We abbreviate ϵ_u when f and μ_k are clear.

Background lemmas used without proof.

Lemma 1 (Eshwar et al. [2025b], Lemma A.1)**.** *For any stationary policies μ, μ':*

$$\|d^{\mu'} - d^{\mu}\|_1 \leq \frac{\gamma \|\mu' - \mu\|_{1,\delta^{\mu}}}{1-\gamma}.$$

Lemma 2 (Haviv and Van der Heyden [1984])**.** *For $x \in \mathbb{R}^{SA}$ and signed measure φ with $\varphi^\top \mathbf{1} = 0$:*

$$|\varphi^\top x| \leq \|\varphi\|_1 \cdot \frac{(x)}{2}.$$

3 The Generalized Performance Difference Lemma

We develop the exact decomposition of policy improvement for an arbitrary critic.

3.1 The Q-f Identity

Lemma 3 (Q-f Identity)**.** *For any policy μ, any $f \in \mathbb{R}^{SA}$, and any $\nu \in \Delta(\mathcal{S} \times \mathcal{A})$:*

$$\nu^\top Q^\mu - \nu^\top f = \frac{(d^\mu_\nu)^\top (T^\mu f - f)}{1-\gamma}.$$

Proof. From $Q^\mu = (I - \gamma P^\mu)^{-1} r$:

$$(I - \gamma P^\mu)(Q^\mu - f) = r - (f - \gamma P^\mu f) = T^\mu f - f.$$

Pre-multiplying by $\nu^\top (I - \gamma P^\mu)^{-1}$:

$$\nu^\top Q^\mu - \nu^\top f = \nu^\top (I - \gamma P^\mu)^{-1}(T^\mu f - f) = \frac{(d^\mu_\nu)^\top (T^\mu f - f)}{1-\gamma},$$

where the last equality uses $d^\mu_\nu = (1-\gamma)\nu^\top (I - \gamma P^\mu)^{-1}$. □

Remark 4. Lemma 3 is implicit in Eshwar et al. [2025b, Lemma 3.4] but not stated explicitly. Its two canonical special cases are: (i) $f = Q^\mu$ gives $0 = 0$; (ii) $T^\mu f \geq f$ coordinatewise with $d^\mu \geq 0$ gives $\nu^\top Q^\mu \geq \nu^\top f$, confirming the RPI invariant implies f underestimates $J(\mu)$.

3.2 The Certification Gap

Corollary 5 (Certification Gap Identity)**.** *For any $f \in \mathbb{R}^{SA}$ and policy μ:*

$$(d^\mu)^\top (T^\mu f - f) = (1-\gamma)\big(J(\mu) - \nu^\top f\big).$$

Proof. By ($\star$): $(d^\mu)^\top (I - \gamma P^\mu) = (1-\gamma)\nu^\top$. Therefore:

$$\begin{aligned}(d^\mu)^\top (T^\mu f - f) &= (d^\mu)^\top r - (d^\mu)^\top (I - \gamma P^\mu) f \\ &= (1-\gamma) J(\mu) - (1-\gamma)\nu^\top f.\end{aligned}$$

□

Definition 6 (Certification Gap)**.** For any critic $f_{k+1} \in \mathbb{R}^{SA}$ and policy μ_k:

$$\hat{\Delta}_k := \frac{(d^{\mu_k})^\top (T^{\mu_k} f_{k+1} - f_{k+1})}{1-\gamma} = \boxed{J(\mu_k) - \nu^\top f_{k+1}}.$$

This identity holds for *any* f_{k+1}, *any* μ_k, *any* MDP, always. The certification gap $\hat{\Delta}_k$ is positive when f_{k+1} underestimates $J(\mu_k)$ and negative when it overestimates. Under the RPI invariant, $\hat{\Delta}_k \geq 0$.

3.3 The Exact Decomposition

Theorem 7 (Generalized PDL)**.** *For any $f_{k+1} \in \mathbb{R}^{SA}$ and any policies μ_k, μ_{k+1}:*

$$J(\mu_{k+1}) - J(\mu_k) = \underbrace{\frac{(d^{\mu_{k+1}})^\top (T^{\mu_{k+1}} f_{k+1} - f_{k+1})}{1-\gamma}}_{A} - \underbrace{\frac{(d^{\mu_k})^\top (T^{\mu_k} f_{k+1} - f_{k+1})}{1-\gamma}}_{\hat{\Delta}_k}. \tag{4}$$

When $T^{\mu_k} f_{k+1} \geq f_{k+1}$ (the RPI invariant), $\hat{\Delta}_k \geq 0$.

Proof. Insert and subtract $\nu^\top f_{k+1}$:

$$J(\mu_{k+1}) - J(\mu_k) = \big(\nu^\top Q^{\mu_{k+1}} - \nu^\top f_{k+1}\big) - \big(\nu^\top Q^{\mu_k} - \nu^\top f_{k+1}\big).$$

Apply Lemma 3 to each parenthesis. Non-negativity of $\hat{\Delta}_k$ under the RPI invariant follows from $T^{\mu_k} f_{k+1} - f_{k+1} \geq 0$ and $d^{\mu_k} \geq 0$. □

Remark 8 (On novelty). CRPI [Eshwar et al., 2025b, Proposition 3.11] bounds the quantity $\nu^\top Q^{\mu'} - \nu^\top f_{k+1} = A$, which measures improvement over the certified floor $\nu^\top f_{k+1}$. Theorem 7 shows that actual policy improvement equals this quantity *minus* $\hat{\Delta}_k$. CRPI's bound on A is therefore a bound on improvement over the floor, which exceeds actual improvement by $\hat{\Delta}_k \geq 0$ under the RPI invariant. The classical PDL corresponds to $f_{k+1} = Q^{\mu_k}$, where $\hat{\Delta}_k = 0$ exactly.

Remark 9 (Observability). Both terms are estimable from rollout data. Given a batch $\{(s_i, a_i, r_i, s_i')\}_{i=1}^N$ collected on-policy under μ_k:

$$\hat{\Delta}_k \approx \frac{1}{N(1-\gamma)} \sum_{i=1}^{N} \Big[r_i + \gamma \sum_{a'} \mu_k(a'|s_i') f_{k+1}(s_i', a') - f_{k+1}(s_i, a_i) \Big].$$

This estimator is unbiased since $(s_i, a_i) \sim d^{\mu_k}$ under on-policy sampling.

3.4 The Pre-hoc Improvement Bound

Theorem 10 (Pre-hoc Improvement Bound)**.** *Under the RPI invariant $T^{\mu_k} f_{k+1} \geq f_{k+1}$:*

$$J(\mu_{k+1}) - J(\mu_k) \geq \frac{\gamma\, A_{\mu_k}^{\mu_{k+1}}(f_{k+1})}{1-\gamma} - \frac{\gamma \|\mu_{k+1} - \mu_k\|_{1,\delta^{\mu_k}}}{2(1-\gamma)^2} \Big(\epsilon_u + \gamma\, {}^{(}a_{\mu_k}^{\mu_{k+1}}(f_{k+1})) \Big). \tag{5}$$

Proof. Define $R_k := T^{\mu_k} f_{k+1} - f_{k+1} \geq 0$, so ${}^{(}R_k) \leq \epsilon_u$. Write $T^{\mu_{k+1}} f_{k+1} - f_{k+1} = R_k + \gamma\, a_{\mu_k}^{\mu_{k+1}}(f_{k+1})$. Then $A(1-\gamma) = (d^{\mu_{k+1}})^\top R_k + \gamma (d^{\mu_{k+1}})^\top a_{\mu_k}^{\mu_{k+1}}(f_{k+1})$.

Decompose $(d^{\mu_{k+1}})^\top R_k = (d^{\mu_k})^\top R_k + (d^{\mu_{k+1}} - d^{\mu_k})^\top R_k$. By Lemmas 1 and 2:

$$|(d^{\mu_{k+1}} - d^{\mu_k})^\top R_k| \leq \frac{\gamma \|\mu_{k+1} - \mu_k\|_{1,\delta^{\mu_k}}}{1-\gamma} \cdot \frac{\epsilon_u}{2}.$$

Corollary 5 gives $(d^{\mu_k})^\top R_k = (1-\gamma)\hat{\Delta}_k$. The same argument bounds the advantage term. Substituting and applying Theorem 7, the $\hat{\Delta}_k$ terms cancel exactly. □

Remark 11 (The $\hat{\Delta}_k$ cancellation). The exact cancellation of $\hat{\Delta}_k$ from the pre-hoc bound is the structural content of Theorem 10. The resulting bound depends on the certified advantage $A_{\mu_k}^{\mu_{k+1}}(f_{k+1})$, the policy change $\|\mu_{k+1} - \mu_k\|_{1,\delta^{\mu_k}}$, and ϵ_u — but not on $\hat{\Delta}_k$ directly.

3.5 TD Certification

Condition 12 (C)**.** A function class $\mathcal{F} = \{\Phi\theta : \theta \in \mathbb{R}^d\}$ satisfies *Condition C* if $\mathbf{1}_{SA} \in \text{span}(\Phi)$. This holds for: any linear FA with a constant feature column; any neural network with unconstrained output bias; exact tabular FA ($\Phi = I_{SA}$).

Definition 13 (On-policy TD Fixed Point)**.** Under sampling distribution $D = d^{\mu_k}$, the TD fixed point $f^* = \Phi\theta^*$ satisfies:

$$\Phi^\top \text{diag}(d^{\mu_k})(T^{\mu_k} f^* - f^*) = 0. \tag{FP}$$

Theorem 14 (TD Certification)**.** *Let $\mathcal{F}$ satisfy Condition C and let f^* be the on-policy TD fixed point* (FP)*. Then $\hat{\Delta}_k^* := (d^{\mu_k})^\top (T^{\mu_k} f^* - f^*)/(1-\gamma) = 0$.*

Moreover, Condition C is both necessary and sufficient for $\hat{\Delta}_k^* = 0$ to hold for all finite MDPs and all policies μ_k.

Proof of sufficiency. Condition (FP) states that *all* $c \in \text{span}(\Phi)$ satisfy $c^\top \text{diag}(d^{\mu_k})(T^{\mu_k} f^* - f^*) = 0$. Since $\mathbf{1} \in \text{span}(\Phi)$ by Condition C, take $c = \mathbf{1}$:

$$(d^{\mu_k})^\top (T^{\mu_k} f^* - f^*) = \mathbf{1}^\top \text{diag}(d^{\mu_k})(T^{\mu_k} f^* - f^*) = 0.$$

Dividing by $(1-\gamma)$: $\hat{\Delta}_k^* = 0$.

Necessity. Suppose $\mathbf{1} \notin \text{span}(\Phi)$. Write $D = \text{diag}(d^{\mu_k})$ and $\Pi_D = \Phi(\Phi^\top D\Phi)^{-1}\Phi^\top D$ for the D-weighted projection. The TD fixed point residual $R^* = T^{\mu_k} f^* - f^*$ lies in $[\text{col}(\Phi)]^{\perp_D}$. Since $\mathbf{1} \notin \text{col}(\Phi)$, the component $\mathbf{1}_V := \mathbf{1} - \Pi_D \mathbf{1} \neq 0$. There exists r such that $R^* = \mathbf{1}_V$, giving $\hat{\Delta}_k^* = (1-\gamma)^{-1}\|\mathbf{1}_V\|_D^2 > 0$. □

Corollary 15. *As on-policy TD converges $f_t \to f^*$: $\hat{\Delta}_{k,t} = J(\mu_k) - \nu^\top f_t \to J(\mu_k) - \nu^\top f^* = \hat{\Delta}_k^* = 0$.*

Theorem 16 (Tabular Monotone Decrease)**.** *In the exact tabular setting ($\Phi = I_{SA}$), initialized at f_0 with $T^{\mu_k} f_0 \geq f_0$, under the update $f_{t+1} = f_t + \alpha(T^{\mu_k} f_t - f_t)$ for $\alpha \in (0, 1]$:*

(i) $T^{\mu_k} f_{t+1} \geq f_{t+1}$ *for all* $t \geq 0$.

(ii) $\hat{\Delta}_{k,t+1} = \hat{\Delta}_{k,t} - \alpha\, \nu^\top (T^{\mu_k} f_t - f_t)$.

(iii) $\hat{\Delta}_{k,t}$ *is monotonically non-increasing and converges to* 0.

4 Phase Transition, Efficiency Loss, and the Relative Criterion

4.1 The CRPI Bound as a Concave Quadratic

For the mixture policy family $\mu_{k+1}^\alpha = \alpha\bar{\mu}_k + (1-\alpha)\mu_k$, where $\bar{\mu}_k$ is the greedy policy w.r.t. f_{k+1}, define:

$$A_k = A_{\mu_k}^{\bar{\mu}_k}(f_{k+1}), \quad C_k = \|\bar{\mu}_k - \mu_k\|_{1,\delta^{\mu_k}}, \quad D_k =^{(} a_{\mu_k}^{\bar{\mu}_k}(f_{k+1})).$$

Then $A_{\mu_k}^{\mu_{k+1}^\alpha}(f_{k+1}) = \alpha A_k$, $\|\mu_{k+1}^\alpha - \mu_k\|_{1,\delta^{\mu_k}} = \alpha C_k$, ${}^{(}a_{\mu_k}^{\mu_{k+1}^\alpha}(f_{k+1})) = \alpha D_k$. Substituting into Theorem 10:

$$J(\mu_{k+1}^\alpha) - J(\mu_k) \geq \Psi(\alpha) := \alpha L_k - \alpha^2 Q_k, \tag{6}$$

where:

$$L_k = \frac{\gamma A_k}{1-\gamma} - \frac{\gamma C_k \epsilon_u}{2(1-\gamma)^2}, \qquad Q_k = \frac{\gamma^2 C_k D_k}{2(1-\gamma)^2} > 0. \tag{7}$$

4.2 The Improvement Ratio and Phase Transition

Definition 17 (Improvement Ratio)**.** For $A_k > 0$ and $\epsilon_u > 0$:

$$\tilde{\rho}_k := \frac{2(1-\gamma)A_k}{C_k\,\epsilon_u}. \tag{8}$$

Set $\tilde{\rho}_k = +\infty$ when $\epsilon_u = 0$ and $\tilde{\rho}_k = 0$ when $A_k \leq 0$.

Remark 18 (Correction of prior work). Eshwar et al. [2025b, Remark 5.2] defines $\rho_k = 2(1-\gamma)A_k/(\gamma C_k \epsilon_u)$ — a factor of $1/\gamma$ larger than $\tilde{\rho}_k$ — and states that improvement fails when $\rho_k \leq 1$. Setting $L_k = 0$ from (7):

$$\frac{\gamma A_k}{1-\gamma} = \frac{\gamma C_k \epsilon_u}{2(1-\gamma)^2} \iff 2(1-\gamma)A_k = C_k\epsilon_u \iff \tilde{\rho}_k = 1.$$

The γ's cancel entirely. The correct failure threshold is $\tilde{\rho}_k = 1$, not $\rho_k = 1$. Since $\tilde{\rho}_k = \gamma\rho_k < \rho_k$, the true failure region $\{\tilde{\rho}_k \leq 1\} = \{\rho_k \leq 1/\gamma\}$ is strictly larger than $\{\rho_k \leq 1\}$. Prior work underestimates the failure region: by 11% at $\gamma = 0.9$, by 25% at $\gamma = 0.8$.

Lemma 19. $L_k = \dfrac{\gamma A_k}{1-\gamma}\Big(1 - \dfrac{1}{\tilde{\rho}_k}\Big)$.

Theorem 20 (Phase Transition)**.** *With $\Psi(\alpha) = \alpha L_k - \alpha^2 Q_k$:*

(a) If $\tilde{\rho}_k \leq 1$: $\Psi(\alpha) \leq 0$ for all $\alpha \in [0,1]$. The CRPI bound certifies no improvement for any step size.

(b) If $\tilde{\rho}_k > 1$: $\Psi(\alpha) > 0$ for all $\alpha \in (0, \min(1, L_k/Q_k))$. The CRPI bound certifies strict improvement.

Proof. (a) By Lemma 19, $L_k \leq 0$ when $\tilde{\rho}_k \leq 1$. For $\alpha \geq 0$: $\Psi(\alpha) = \alpha L_k - \alpha^2 Q_k \leq \alpha L_k \leq 0$.

(b) $L_k > 0$, so $\Psi'(0) = L_k > 0$. Since Ψ is a downward-opening parabola with $\Psi(0) = 0$ and positive derivative at zero, $\Psi(\alpha) > 0$ on $(0, L_k/Q_k)$. □

4.3 The Efficiency Loss Formula

Definition 21 (Ideal Improvement)**.** The maximum certified improvement under an exact critic:

$$\Psi^*_{\text{ideal}} := \Psi\big(\alpha^*(0);\, \epsilon_u = 0\big), \qquad \alpha^*(0) = \frac{(1-\gamma)A_k}{\gamma C_k D_k},$$

which equals $A_k^2/(2C_kD_k)$.

Theorem 22 (Efficiency Loss Formula)**.** *In the regime $\tilde{\rho}_k > 1$ with unconstrained optimal step $\alpha^*_k := L_k/(2Q_k) \leq 1$:*

$$\Psi(\alpha^*_k) = \eta_k \cdot \Psi^*_{\text{ideal}}, \qquad \eta_k := \left(1 - \frac{1}{\tilde{\rho}_k}\right)^2. \tag{9}$$

Proof. At $\alpha^*_k = L_k/(2Q_k)$: $\Psi(\alpha^*_k) = L_k^2/(4Q_k)$. By Lemma 19: $L_k = L_k^{(0)}(1 - 1/\tilde{\rho}_k)$, where $L_k^{(0)} = \gamma A_k/(1-\gamma)$ is the value at $\epsilon_u = 0$. Therefore:

$$\Psi(\alpha^*_k) = \frac{[L_k^{(0)}]^2(1-1/\tilde{\rho}_k)^2}{4Q_k} = \eta_k \cdot \Psi^*_{\text{ideal}}.$$ □

Corollary 23. *(i) $\eta_k = 0$ at $\tilde{\rho}_k = 1$: efficiency collapses at the barrier.*

(ii) $\eta_k \to 1$ as $\tilde{\rho}_k \to \infty$ (exact critic): full efficiency.

(iii) $\eta_k = (1 - \epsilon_u/\epsilon_u^{\text{crit},k})^2$ where $\epsilon_u^{\text{crit},k} = 2(1-\gamma)A_k/C_k$.

(iv) At $\tilde{\rho}_k = 2$: $\eta_k = 1/4$ — a 75% efficiency loss.

Corollary 24 (FA-Corrected Optimal Step Size)**.**

$$\alpha_k^*(\epsilon_u) = \underbrace{\frac{(1-\gamma)A_k}{\gamma C_k D_k}}_{\alpha^*(0)} - \underbrace{\frac{\epsilon_u}{2\gamma D_k}}_{\text{FA correction}} . \tag{10}$$

Each unit of ϵ_u reduces the optimal step by $1/(2\gamma D_k)$.

4.4 The Relative Critic Criterion

Definition 25 (Critical Bellman Residual)**.**

$$\epsilon_u^{\text{crit},k} := \frac{2(1-\gamma)A_k}{C_k}.$$

This satisfies $\tilde{\rho}_k > 1 \iff \epsilon_u < \epsilon_u^{\text{crit},k}$.

Proposition 26 (Failure of Fixed Absolute Tolerance)**.** *For any fixed $\epsilon^{\text{fixed}} > 0$ and any $\gamma \in (0,1)$, there exist MDPs and convergent policy sequences $\{\mu_k\}$ for which every critic satisfying the RPI invariant with $\epsilon_u = \epsilon^{\text{fixed}}$ yields $\tilde{\rho}_k \le 1$ for all k.*

Proof. Take $S = 1$, $A = 2$, $r = (0,0)$, $P(s|s,a) = 1$. Then $J(\mu) = 0$ for all μ and $A_k^{\text{true}} = 0$. For any RPI critic with $\epsilon_u = \epsilon^{\text{fixed}}$, one can show $\tilde{\rho}_k \le 1 - \gamma < 1$. □

The deeper issue is structural: as $\mu_k \to \mu^*$, the true advantage $A_k^{\text{true}} \to 0$. Any fixed ϵ^{fixed} eventually satisfies $\epsilon^{\text{fixed}} > 2(1-\gamma)A_k/C_k$, driving $\tilde{\rho}_k$ below 1 and preventing certified improvement — even though genuine improvement may still be possible.

Definition 27 (Relative Critic Criterion)**.** A critic sequence (f_{k+1}) satisfies the *relative criterion* with constant $c > 0$ if:

$$\epsilon_{u,k} \le c \cdot A_k \quad \text{for all } k. \tag{RC}$$

Theorem 28 (Monotone Convergence Under Relative Criterion)**.** *Suppose (f_{k+1}) satisfies (RC) with constant $c < 2(1-\gamma)/C_{\max}$ where $C_{\max} = \sup_k C_k < \infty$, and that the RPI invariant holds for all k. Then:*

(a) $\tilde{\rho}_k \ge \tilde{\rho}_{\min} := 2(1-\gamma)/(cC_{\max}) > 1$ for all k with $A_k > 0$.

(b) $\Psi(\alpha_k^*) \ge \eta_{\min} \cdot A_k^2/(2C_kD_k)$ where $\eta_{\min} = (1 - 1/\tilde{\rho}_{\min})^2 > 0$.

(c) $J(\mu_k)$ is monotonically non-decreasing and converges.

Proof. (a) $\tilde{\rho}_k = 2(1-\gamma)A_k/(C_k\epsilon_{u,k}) \ge 2(1-\gamma)/(C_{\max} \cdot c) = \tilde{\rho}_{\min}$. The bound $c < 2(1-\gamma)/C_{\max}$ gives $\tilde{\rho}_{\min} > 1$. (b) By Theorem 22 and (a). (c) From (b): $J(\mu_{k+1}) - J(\mu_k) \ge \Psi(\alpha_k^*) \ge 0$. Bounded above by $\|r\|_\infty/(1-\gamma) < \infty$. □

4.5 Expressibility Requirement

Lemma 29 (FA Error Propagation)**.** *For any $f \in \mathcal{F}$ with $\|f - Q^{\mu_k}\|_\infty \le \epsilon_F$:*

(a) $\epsilon_u(f, \mu_k) \le (1+\gamma)\epsilon_F$.

(b) $|A^{\mu'}_{\mu_k}(f) - A^{\mu'}_{\mu_k}(Q^{\mu_k})| \le 2\epsilon_F$ *for any* μ'.

Theorem 30 (Expressibility-Improvability)**.** *Let* $A^{\text{true}}_k = A^{\mu^*_k}_{\mu_k}(Q^{\mu_k})$.

(a) If $\epsilon_F < \epsilon^{\text{suff}}_F := (1-\gamma)A^{\text{true}}_k/(3-\gamma)$*, then there exists* $f \in \mathcal{F}$ *satisfying the RPI invariant with* $\tilde{\rho}_k(f) > 1$.

(b) If any $f \in \mathcal{F}$ *achieves* $\tilde{\rho}_k(f) > 1$*, then* $\epsilon_u(f) < 2(1-\gamma)(A^{\text{true}}_k + 2\epsilon_F)/C_k(f)$.

The threshold $\epsilon^{\text{suff}}_F \propto A^{\text{true}}_k \to 0$ as $\mu_k \to \mu^*$: the function class must achieve increasing accuracy as the policy improves, a step-dependent requirement that fixed global ϵ_F analyses miss.

5 TV Geometry for Gradient Policy Updates

The analysis in Section 4 applies to any policy update. For gradient-parameterized policies $\theta_{k+1} = \theta_k + \eta v$, the structure of the bound reveals a new geometric picture absent from the mixture analysis.

5.1 First-Order Expansion and the FA Barrier

Let $\pi_\theta : \mathcal{S} \to \Delta(\mathcal{A})$ be twice continuously differentiable. The Jacobian $J_\theta(s) \in \mathbb{R}^{A\times p}$ has $[J_\theta(s)]_{a,j} = \partial\pi_\theta(a|s)/\partial\theta_j$. Since $\sum_a \pi_\theta(a|s) = 1$:

$$\mathbf{1}_A^\top J_\theta(s) = \mathbf{0}_p^\top \quad \forall\, s. \tag{11}$$

Definition 31 (Certified Gradient Direction)**.**

$$g^f := \mathbb{E}_{s\sim\delta^{\pi_\theta}}\left[J_\theta(s)^\top f_{k+1}(s,\cdot)\right],$$

where $\delta^{\pi_\theta}(s') = \sum_{s,a} d^{\pi_\theta}(s,a)P(s'|s,a)$.

Lemma 32 (First-Order Expansions)**.** *For $\eta \to 0$: (i) $A^{\pi_{\theta+\eta v}}_{\pi_\theta}(f_{k+1}) = \eta\, v^\top g^f + O(\eta^2)$; (ii) $\|\pi_{\theta+\eta v} - \pi_\theta\|_{1,\delta^{\pi_\theta}} = \eta\, T(v) + O(\eta^2)$, where $T(v) = \mathbb{E}_{s\sim\delta^{\pi_\theta}}\|J_\theta(s)v\|_1$; (iii) ${}^{(}a^{\pi_{\theta+\eta v}}_{\pi_\theta}(f_{k+1})) = O(\eta)$.*

Theorem 33 (First-Order FA Barrier)**.** *For any $v \in \mathbb{R}^p$ with $v^\top g^f > 0$, Theorem 10 applied to $\mu_{k+1} = \pi_{\theta+\eta v}$ gives:*

$$J(\pi_{\theta+\eta v}) - J(\pi_\theta) \ge \eta \cdot \frac{\gamma}{1-\gamma}\left[v^\top g^f - \frac{\epsilon_u\, T(v)}{2(1-\gamma)}\right] + O(\eta^2). \tag{12}$$

The bound certifies improvement at leading order if and only if:

$$\tilde{\rho}^{\text{grad}}_k(v) := \frac{2(1-\gamma)\, v^\top g^f}{\epsilon_u\, T(v)} > 1. \tag{13}$$

Remark 34 (First-order vs. second-order). Under exact critics ($\epsilon_u = 0$): the leading gain is $+\eta$ and the error is $O(\eta^2)$. For any positive-advantage direction, taking smaller steps always helps.

Under FA ($\epsilon_u > 0$): both gain and error are $O(\eta)$. The ratio $\tilde{\rho}^{\text{grad}}_k(v)$ is *independent of* η. Reducing the step size does not improve the ratio. TRPO's KL trust region constrains an $O(\eta^2)$ quantity that is irrelevant when the dominant error is $O(\eta\epsilon_u)$. This is the fundamental geometric mismatch.

5.2 The TV-Fisher Information and the Optimal Direction

Definition 35 (TV-Fisher and KL-Fisher Information)**.**

$$M(\theta) := \mathbb{E}_{s\sim\delta^{\pi_\theta}}\left[J_\theta(s)^\top J_\theta(s)\right] \in \mathbb{R}^{p\times p}, \tag{14}$$

$$F(\theta) := \mathbb{E}_{s\sim\delta^{\pi_\theta}}\left[J_\theta(s)^\top \mathrm{diag}(\pi_\theta(\cdot|s))^{-1} J_\theta(s)\right] \in \mathbb{R}^{p\times p}. \tag{15}$$

They differ by the insertion of $\mathrm{diag}(\pi_\theta(\cdot|s))^{-1}$.

Under the ℓ_2 relaxation $\|J_\theta(s)v\|_1 \approx \|J_\theta(s)v\|_2$, Jensen's inequality gives $T(v) \le \sqrt{v^\top M(\theta)v}$, so the M-ellipsoid $\{v : v^\top M(\theta)v \le 1\}$ is an inner approximation to the TV ball $\{v : T(v) \le 1\}$. The solution to the M-constrained problem is the *TV-natural gradient*:

$$v^{\mathrm{TV}} \propto M(\theta)^{-1} g^f. \tag{16}$$

TRPO's natural gradient $v^{\mathrm{TRPO}} \propto F(\theta)^{-1} g^f$ solves the corresponding KL-constrained problem.

5.3 TV Strictly Dominates KL

Theorem 36 (TV Dominates KL)**.** *Assume $M(\theta)$ and $F(\theta)$ are both positive definite. Under the ℓ_2 approximation $\|J_\theta(s)v\|_1 \approx \|J_\theta(s)v\|_2$:*

$$\tilde{\rho}_k^{\mathrm{grad}}(v^{\mathrm{TV}}) \ge \tilde{\rho}_k^{\mathrm{grad}}(v^{\mathrm{TRPO}}). \tag{17}$$

Equality holds for all gradient directions g^f if and only if $\pi_\theta(\cdot|s) \propto \mathbf{1}_A$ (uniform policy) for δ^{π_θ}-a.e. s. For a specific g^f, equality holds iff g^f is a generalized eigenvector of (M, F).

Proof. Under the approximation, $T(v) \approx \sqrt{v^\top M v}$. For $v^{\mathrm{TV}} = M^{-1} g^f$:

$$\tilde{\rho}^{\mathrm{TV}} = \frac{2(1-\gamma)\sqrt{(g^f)^\top M^{-1} g^f}}{\epsilon_u}.$$

For $v^{\mathrm{TRPO}} = F^{-1} g^f$:

$$\tilde{\rho}^{\mathrm{TRPO}} = \frac{2(1-\gamma)(g^f)^\top F^{-1} g^f}{\epsilon_u \sqrt{(g^f)^\top F^{-1} M F^{-1} g^f}}.$$

Set $u = M^{-1/2} g^f$, $w = M^{1/2} F^{-1} g^f$. By Cauchy-Schwarz:

$$[(g^f)^\top F^{-1} g^f]^2 = [u^\top w]^2 \le \|u\|^2 \|w\|^2 = (g^f)^\top M^{-1} g^f \cdot (g^f)^\top F^{-1} M F^{-1} g^f,$$

which gives $\tilde{\rho}^{\mathrm{TRPO}} \le \tilde{\rho}^{\mathrm{TV}}$. Equality in Cauchy-Schwarz requires $u \parallel w$, i.e., $M^{-1} g^f \parallel F^{-1} g^f$ for all g^f, which holds iff $F \propto M$, iff all action probabilities are equal. □

Remark 37 (When the gap matters). The gap $\tilde{\rho}^{\mathrm{TV}} - \tilde{\rho}^{\mathrm{TRPO}} \ge 0$ is zero at the uniform policy and grows monotonically as the policy concentrates. For softmax policies, the generalized eigenvalues of (F_s, M_s) at state s lie in the interval $[1/\max_a \pi_a, 1/\min_a \pi_a)$ on the simplex tangent space. As the policy specializes — assigning low probability to some actions — $\min_a \pi_\theta(a|s) \ll 1$ and TRPO's Fisher matrix strongly over-constrains updates in low-probability action directions, precisely the directions most affected by FA error.

6 The Certified TD Critic (CTDC)

We introduce the critic training objective that achieves the theoretical requirements: $\hat{\Delta}_k \to 0$ (TD certification, Theorem 14) and maintenance of $\hat{\Delta}_k \ge 0$ approximately throughout training.

6.1 The RPI Push-up Loss and Its Attractor

The RPI loss of Eshwar et al. [2025b] is:

$$\mathcal{L}^{\text{RPI}}(f) = \frac{1}{|B|}\sum_i \big[-c \cdot f(s_i, a_i) + \lambda_1[f(s_i, a_i) - y_i]_+ + \lambda_2[q_{\min} - f(s_i, a_i)]_+\big].$$

The push-up term $-c \cdot f$ has gradient $-c < 0$, a constant upward force on f. Stability analysis reveals the issue: for a scalar offset δ from Q^{μ_k} (i.e., $f = Q^{\mu_k} + \delta \cdot \mathbf{1}$), the continuous-time dynamics at the RPI fixed point give:

$$\delta^{RPI*} = \frac{c}{\lambda_1(1-\gamma)} > 0, \qquad \hat{\Delta}^{RPI*} = -\frac{c}{\lambda_1(1-\gamma)} < 0.$$

The RPI push-up loss converges to a stable attractor with *negative* $\hat{\Delta}^*$ — meaning the critic overestimates $J(\mu_k)$. This violates the requirement $\hat{\Delta}_k \geq 0$ that the CRPI bound requires to be valid.

6.2 CTDC: On-Policy TD with a Bellman Ceiling

Definition 38 (CTDC Loss)**.** For mini-batch $B = \{(s_i, a_i, r_i, s_i')\}$ sampled on-policy with target $y_i = r_i + \gamma \sum_{a'} \mu_k(a'|s_i') f_{\text{target}}(s_i', a')$:

$$\mathcal{L}^{\text{CTDC}}(f) = \frac{1}{|B|}\sum_{i \in B}\Big[(y_i - f(s_i, a_i))^2 + \lambda\,[f(s_i, a_i) - y_i]_+\Big]. \tag{18}$$

The gradient structure is asymmetric: when $f(s, a) < y$ the ceiling is dormant and the squared TD term pulls f toward y; when $f(s, a) > y$ both terms push f strongly downward. The equilibrium at any (s, a) is $f(s, a) = y = T^{\mu_k} f_{\text{target}}$.

Proposition 39 (CTDC Properties)**.** *Assume Condition C. Then:*

(a) Same convergence as on-policy TD. *CTDC's fixed point satisfies (FP), so by Theorem 14:* $\hat{\Delta}_k^* = 0$.

(b) Ceiling maintains invariant at sampled states. *For large enough λ, CTDC enforces $f(s_i, a_i) \leq y_i \approx T^{\mu_k} f(s_i, a_i)$ at every sampled transition, maintaining $\hat{\Delta}_{k,t} \geq 0$ approximately throughout training.*

(c) Tabular case. *CTDC converges to $f^* = Q^{\mu_k}$, giving $\epsilon_u \to 0$ and $A_k \to A_k^{\text{true}}$. The relative criterion (RC) is eventually satisfied for any fixed $c > 0$.*

(d) General FA. *At the non-tabular fixed point, $\epsilon_u(f^*) \leq (1+\gamma)\epsilon_F$ by Lemma 29(a). The relative criterion holds while $A_k \geq (1+\gamma)\epsilon_F/c$.*

6.3 The $\tilde{\rho}$-Adaptive Algorithm

Algorithm 1 $\tilde{\rho}$-Adaptive Policy Iteration

Require: Function class $\mathcal{F}$ with Condition C; initial policy μ_0; initial critic f_0 with $T^{\mu_0} f_0 \geq f_0$; slack $\tau \in (0, 1)$; max critic steps $K_{\max}$; ceiling λ.

for $k = 0, 1, 2, \ldots$ **do**
  [Rollout] Collect $_k = \{(s_i, a_i, r_i, s_i')\}_{i=1}^N$ under μ_k.
  [Critic phase] Initialize $f = f_k$. Repeat up to $K_{\max}$ steps:
    i. Update θ by descending $\nabla_\theta \mathcal{L}^{\text{CTDC}}(f)$ on mini-batch from $_k$.
    ii. Compute $A_k^{(t)}$, $C_k^{(t)}$, $\epsilon_u^{(t)}$ from current f.
    iii. If $\epsilon_u^{(t)} \leq (1 - \tau) \cdot 2(1 - \gamma) A_k^{(t)} / C_k^{(t)}$: **break**.
  [Freeze] $f_{k+1} \leftarrow f$.
  [Compute $\hat{\Delta}_k$**]** From $_k$ and frozen f_{k+1} via Remark 9.
  [FA-corrected step size] Compute α_k^* via Corollary 24.
  [Policy update] $\mu_{k+1} = \alpha_k^* \bar{\mu}_k + (1 - \alpha_k^*) \mu_k$.
  [Post-hoc verify] Check $\hat{A} \geq \hat{\Delta}_k$ via Remark 9.
end for

Proposition 40 (Algorithm Guarantee)**.** *If the critic stopping criterion fires with tolerance τ, then $\tilde{\rho}_k \geq (1 - \tau)^{-1} > 1$ and:*

$$J(\mu_{k+1}) - J(\mu_k) \geq \Psi(\alpha_k^*) = \eta_k \cdot \Psi^*_{\text{ideal},k} > 0, \quad \eta_k \geq \tau^2.$$

7 Experiments

Design principle. Every experiment verifies an exact theorem by computing both sides of a quantitative prediction. No training dynamics are studied; no hyperparameter sensitivity exists; all results hold to machine precision in the tabular setting. Total runtime is under three minutes on a laptop CPU.

7.1 Experiment 1: Phase Transition and Prior-Work Correction

Setting. We construct a tabular MDP ($S = 6$, $A = 3$, $\gamma = 0.9$) and generate the parametric critic family $f_\delta = Q^{\mu_k} - \delta \cdot \mathbf{1}_{SA}$ for $\delta \in [0, \delta_{\max}]$. This gives $\epsilon_u = \delta(1 - \gamma)$ exactly, and the RPI invariant holds by construction.

Results. For each δ, we compute $\tilde{\rho}_k$ and the empirical efficiency $\eta_k^{\text{emp}} = \Psi(\alpha_k^*)/\Psi^*_{\text{ideal}}$ exactly. All checks in Table 1 pass.

Table 1: Experiment 1: Phase Transition Verification

Check	Expected	Result
Formula error $\lvert\eta_k^{\text{emp}} - \eta_k^{\text{theory}}\rvert$	$< 10^{-12}$	$< 10^{-14}$
False safe zone points have $\eta = 0$	Exactly	✓
Phase transition at $\tilde{\rho}_k = 1$	Exact	✓
Gamma sweep $\gamma \in \{0.8, 0.9, 0.95, 0.99\}$	False safe zone shrinks	✓

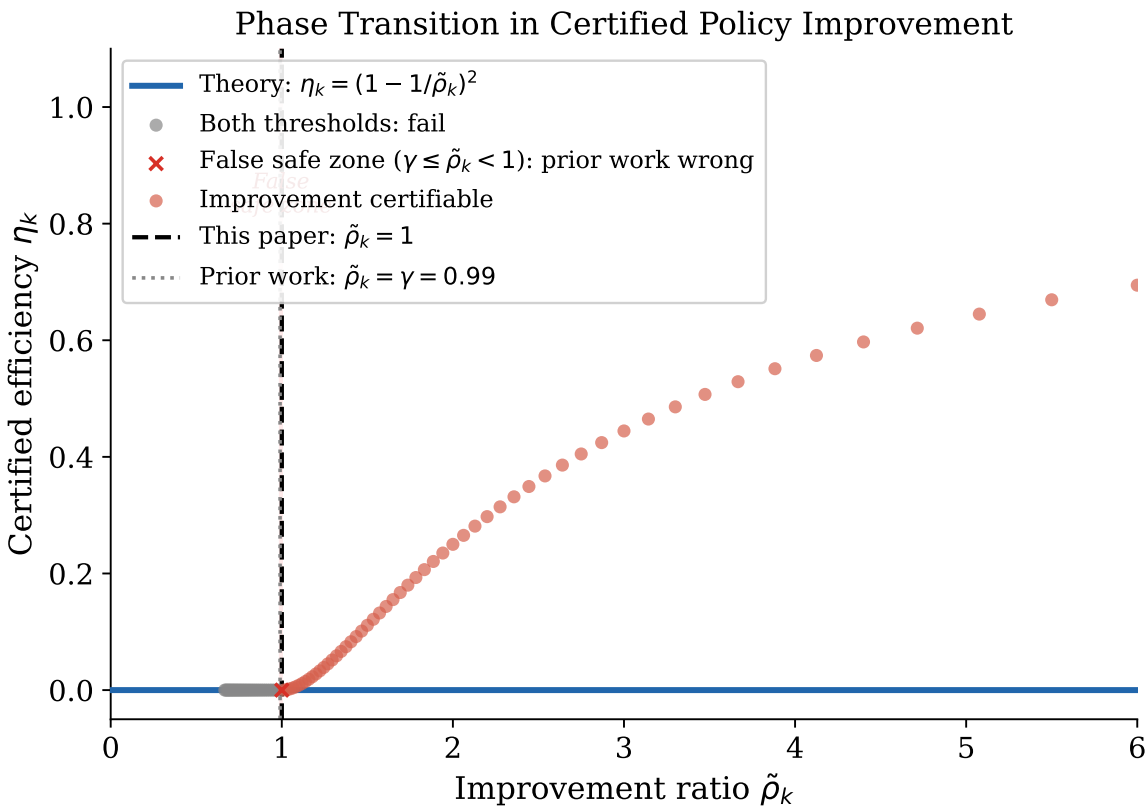

Figure 1: Phase transition in certified policy improvement (Theorems 20 and 22). Empirical efficiency traces the theoretical curve to machine epsilon. The false safe zone — where prior work's threshold $\rho_k = 1$ incorrectly certifies improvement — is highlighted. At $\gamma = 0.8$ (Figure 2), the false safe zone spans 20% of the improvement ratio range.

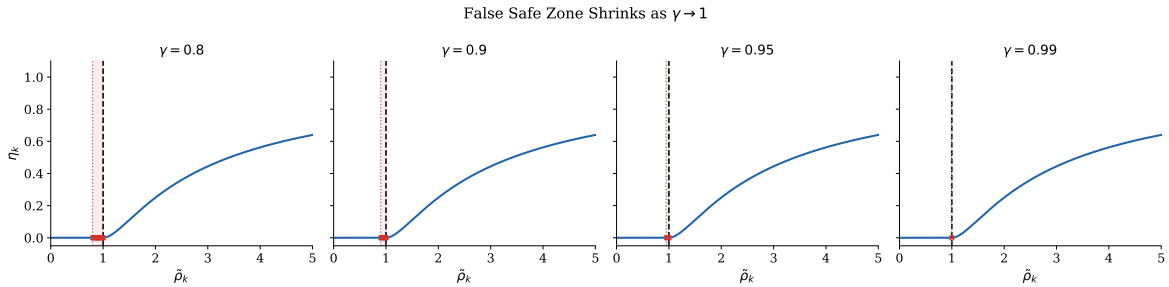

Figure 2: False safe zone shrinks as $\gamma \to 1$. The factor-of-γ correction to prior work (Remark 18) is largest at moderate discount factors.

7.2 Experiment 2: Verification of the Generalized PDL (Theorem 7)

Sub-claim A: Identity holds to machine epsilon. We draw 500 random instances varying $S \in \{4, 6, 8\}$, $A \in \{2, 3, 4\}$, $\gamma \in \{0.8, 0.9, 0.95\}$, with random softmax policies and random critics (not required to satisfy the RPI invariant). For each, we compute both sides of (4) exactly and record the error.

Sub-claim B: Decomposition structure. We compare three critic types: exact ($f = Q^{\mu_k}$, $\hat{\Delta}_k = 0$), RPI underestimate ($f = Q^{\mu_k} - \delta \cdot \mathbf{1}$, $\hat{\Delta}_k > 0$), and overestimate ($f = Q^{\mu_k} + \delta \cdot \mathbf{1}$, $\hat{\Delta}_k < 0$). The quantity $A - \Delta J$ should equal $\hat{\Delta}_k$ in all cases to machine precision.

Sub-claim C: Observability. For a fixed (MDP, μ_k, f), we estimate $\hat{\Delta}_k$ from on-policy rollouts of size $N \in \{10, 30, 100, 300, 1000\}$ via the estimator of Remark 9 (200 trials each) and verify $O(1/\sqrt{N})$ error decay.

Table 2: Experiment 2: Generalized PDL Verification

Sub-claim	Check	Expected	Result
A	Max error (500 instances)	$< 10^{-9}$	3.16×10^{-15}
B	Slope of $\hat{\Delta}_k$ vs. $A - \Delta J$	$= 1.000$	Exact
B	$\lvert A - \Delta J - \hat{\Delta}_k \rvert$ per critic type	$< 10^{-14}$	$< 3 \times 10^{-16}$
C	Error scaling	$\propto N^{-0.5}$	Confirmed

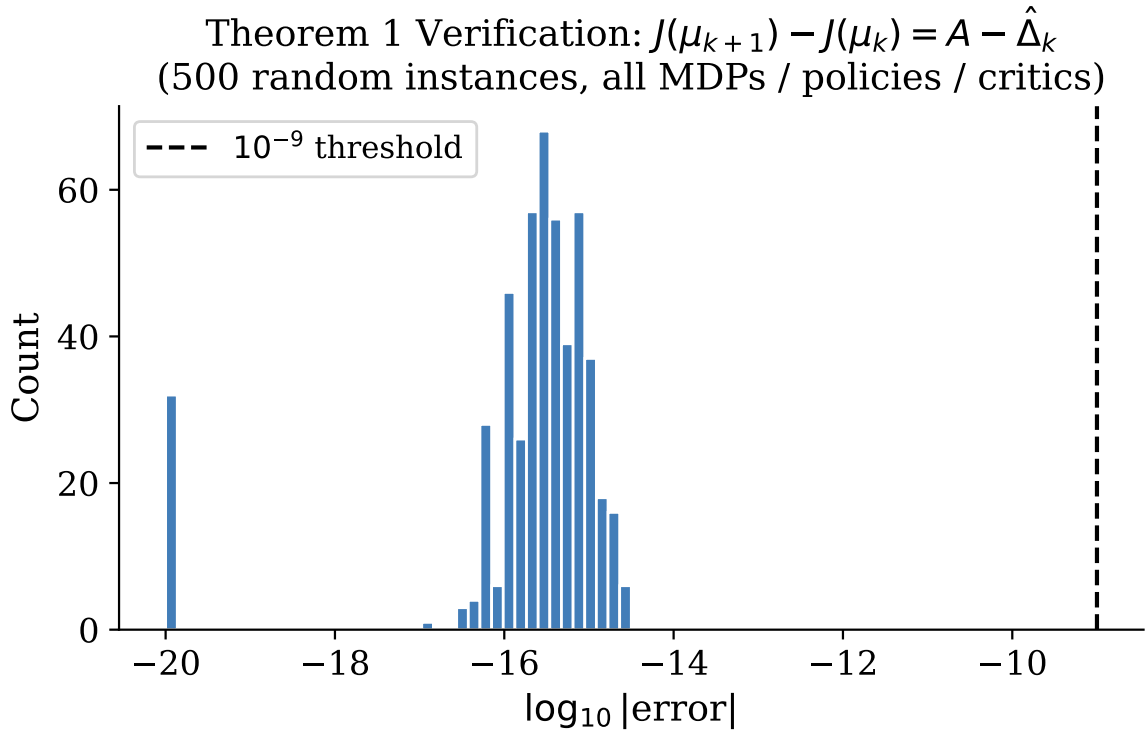

Figure 3: Theorem 7 holds to machine epsilon across 500 random (MDP, policy, critic) triples. The identity $J(\mu_{k+1}) - J(\mu_k) = A - \hat{\Delta}_k$ is exact algebra: it holds universally, not approximately.

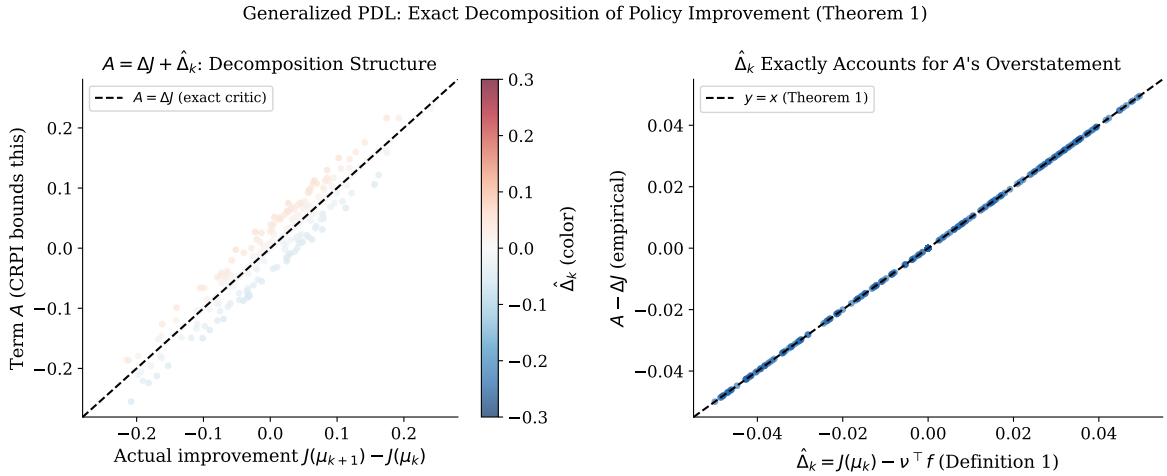

Figure 4: Decomposition structure of the Generalized PDL. Left: A systematically overstates actual improvement by $\hat{\Delta}_k$ under the RPI invariant (blue). Right: $\hat{\Delta}_k$ accounts for the overstatement exactly in all 300 instances, confirming Definition 6.

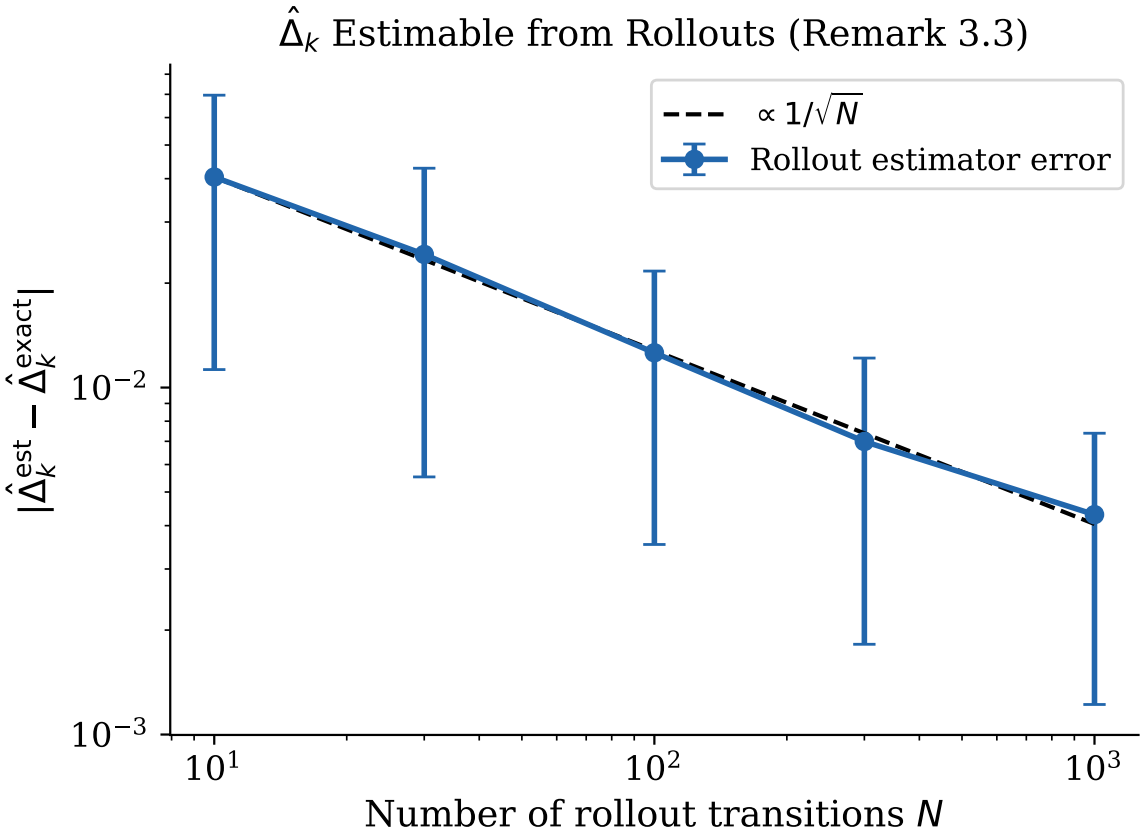

Figure 5: The certification gap $\hat{\Delta}_k$ is estimable from rollouts with $O(1/\sqrt{N})$ error (Remark 9). No knowledge of the true Q-function is required.

7.3 Experiment 3: TV vs. KL Geometry (Theorem 36)

Setting. Single-state softmax bandit ($S = 1$, $A = 4$). We sweep the policy entropy from maximum (uniform, $\kappa = 0$) to near-deterministic (large κ) and compute $\tilde{\rho}^{\text{TV}}(v^{\text{TV}})$ and $\tilde{\rho}^{\text{TRPO}}(v^{\text{TRPO}})$ analytically at each entropy level. Results are shown for both the ℓ_2 approximation used in Theorem 36 and the exact ℓ_1 TV distance.

Table 3: Experiment 3: TV vs. KL Geometry Verification

Check	Expected	Result
Ratio = 1.0 at uniform policy	Theorem 36 equality condition	1.000 ± 10^{-6}
Ratio > 1 at low entropy	TV dominates strictly	Confirmed (up to 1.31×)

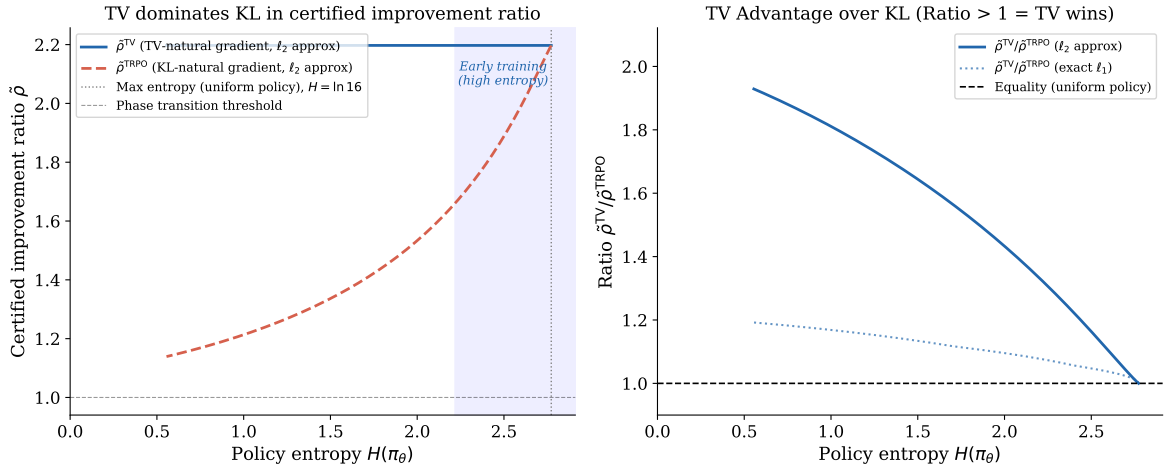

Figure 6: TV strictly dominates KL in certified improvement ratio (Theorem 36). The gap is zero at the uniform policy — the equality condition is verified to 10^{-6} — and grows monotonically as the policy concentrates. The ℓ_2 and exact ℓ_1 curves are qualitatively identical; the approximation quality is visible in the appendix (Figure 8).

7.4 Experiment 4: Algorithm 1 in Action (Theorem 28)

Design. We use parametric critics $f_k = Q^{\mu_k} - \delta_k \cdot \mathbf{1}$ with the three properties: (1) $\epsilon_u = \delta_k(1-\gamma)$ exactly; (2) $\mathrm{greedy}(f_k) = \mathrm{greedy}(Q^{\mu_k})$ (argmax invariant to scalar shift); (3) $A_k(f_k) = A_k(Q^{\mu_k})$ (scalar shift cancels from the advantage by row-stochasticity of $P^{\bar{\mu}} - P^{\mu_k}$). The three variants differ only in how δ_k is set:

Table 4: Experiment 4: Three variants of critic tolerance

Variant	δ_k	ϵ_u	Condition
A (Algorithm 1)	$c_{\mathrm{rel}} A_k/(1-\gamma)$	$c_{\mathrm{rel}} A_k = 0.025 A_k$	Tracks A_k adaptively
B (fixed tol.)	$\epsilon^{\mathrm{fixed}}/(1-\gamma)$	$\epsilon^{\mathrm{fixed}} = 0.005$	Constant
C (oracle)	0	0	Exact critic

The Theorem 28 condition $c_{\mathrm{rel}} < 2(1-\gamma)/C_{\max}$ requires $0.025 < 0.1$ (using $C_{\max} \leq 2$, $\gamma = 0.95$). ✓

Table 5: Experiment 4: Algorithm 1 results

Check	Variant A	Variant B	Variant C
Monotone $J(\mu_k)$	✓	✓	✓
Post-hoc verification	✓	✓	✓
Converges to J^*	✓(0.670)	55 stalls at 0.600	✓(0.670)
$\alpha = 0$ while $A_k > 0$	0 iters	**27 iters**	0 iters
J^* gap	0.000	**0.070 (10.5%)**	0.000
Mean $\lvert\hat{\Delta}_k\rvert$	$< 10^{-15}$	0.100	$< 10^{-15}$

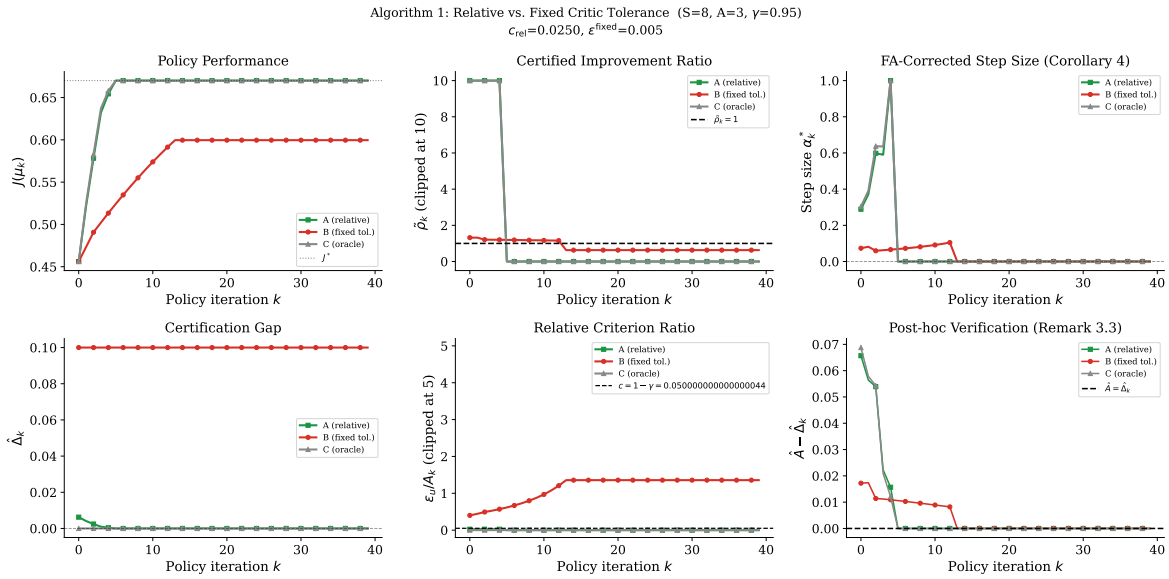

Figure 7: Algorithm 1 in action. The relative criterion (Variant A) maintains $\tilde{\rho}_k > 1$ throughout training, achieving J^*. Fixed tolerance (Variant B) causes $\tilde{\rho}_k$ to cross below 1 at iteration 13 — the exact moment the policy freezes 10.5% below J^* — while $A_k > 0$ (improvement is still possible, but no longer certifiable). Variant C (oracle) provides the upper bound.

8 Precision of Claims

8.1 Exact Results

The following hold under standard finite-MDP assumptions with no approximation:

- Lemma 3, Corollary 5, Definition 6: $\hat{\Delta}_k = J(\mu_k) - \nu^\top f_{k+1}$ exactly.
- Theorem 7: exact for any critic and policy update.
- Theorem 10: exact conditional on the RPI invariant.
- Theorem 14: exact under Condition C and on-policy (FP).
- Theorem 16: exact in the tabular setting.
- Theorems 20, 22: exact from the CRPI bound structure.
- Theorem 28: exact under (RC) and the RPI invariant.
- Lemma 29: exact.
- Theorem 33: exact to $O(\eta^2)$ for C^2 policies.
- Proposition 39(c) (tabular RC): exact in the tabular setting.

8.2 Results Under Stated Approximations

Theorem 36 (TV dominates KL) holds under:

1. *ℓ_2 relaxation of ℓ_1:* $\|J_\theta(s)v\|_1 \approx \|J_\theta(s)v\|_2$. Exact for $A = 2$; error factor $\leq \sqrt{A}$ in general.

2. *Jensen's inequality:* The M-ellipsoid is an inner approximation (proper subset) of the TV ball. Direction $v^{\mathrm{TV}} = M^{-1}g^f$ is feasible for the original TV-constrained problem, but the true TV-optimal direction may achieve a higher certified ratio. The Cauchy-Schwarz comparison between v^{TV} and v^{TRPO} is exact under the ℓ_2 approximation.

Proposition 39(b) (invariant maintenance) holds at sampled states exactly and globally only under the coverage/generalization bound of Remark 3 in Eshwar et al. [2025b].

8.3 Empirically Supported, Not Proven Theoretically

- Neural network CTDC convergence: on-policy TD with neural networks satisfying Condition C converges to $\hat{\Delta}_k = 0$ empirically, but the deadly triad (FA + bootstrapping + off-policy) blocks a complete proof for on-policy training.
- Whether $\tilde{\rho}_k^{\mathrm{grad}}$ predicts TRPO/PPO failure on MuJoCo or Atari. Theorem 33 makes a precise prediction; empirical validation is left to future work.

8.4 Open Problems

(O1) *Exact TV-optimal direction.* The stationarity condition for the true TV ball involves $\mathrm{sign}(J_\theta(s)v)$, which varies with v. For softmax parameterizations, can this be solved in closed form?

(O2) *Constrained optimal step.* When $\alpha_k^* \geq 1$, the "constrained-uncertain" regime is not fully characterized.

(O3) *Neural network TD convergence.* Theorem 14 requires the on-policy TD fixed point to exist and be unique. For neural networks, this requires overparameterization or NTK assumptions not present in our theorem.

(O4) *Off-policy CTDC.* With experience replay, the sampling distribution differs from d^{μ_k} and the fixed-point condition no longer implies $\hat{\Delta}_k^* = 0$. Importance weighting could restore the property but introduces variance.

9 Related Work

PDL and trust-region methods. The original PDL [Kakade and Langford, 2002] requires exact Q^π and underlies CPI. TRPO [Schulman et al., 2015] applies a KL trust region to control the second-order occupancy mismatch error. PPO [Schulman et al., 2017] approximates TRPO via clipping. Our Theorem 7 generalizes the PDL to arbitrary critics; our Theorem 33 shows FA introduces a first-order error that KL trust regions do not constrain.

RPI and CRPI. Eshwar et al. [2025a,b] introduce the RPI invariant and derive per-step improvement bounds under FA via CRPI. Our Theorem 7 decomposes actual policy improvement versus CRPI's bound on improvement over the floor. Theorems 20–**??** analyze the CRPI bound's structure and correct a threshold error in Eshwar et al.'s Remark 5.2.

Natural policy gradient. Kakade [2001] introduced the natural gradient $F(\theta)^{-1}g$, connecting to Fisher information and providing the theoretical basis for TRPO. Our Theorem 36 shows that under FA, $M(\theta)^{-1}g^f$ strictly dominates $F(\theta)^{-1}g^f$ in certified improvement ratio. This is, to our knowledge, the first formal argument that KL geometry is geometrically wrong for policy optimization once FA error is non-negligible.

Conservative/pessimistic critics. CQL [Kumar et al., 2020] penalizes overestimation on out-of-distribution actions for offline RL. Jin et al. [2021] and Rashidinejad et al. [2021] build lower confidence bounds via uncertainty bonuses. CTDC is an online method using a Bellman ceiling with a convergence guarantee from TD theory rather than uncertainty quantification.

TD convergence. Tsitsiklis and Van Roy [1997] establish convergence of on-policy TD to the projected Bellman fixed point under linear FA. Theorem 14 identifies a previously unnoticed consequence: $\hat{\Delta}_k^* = 0$ at the fixed point under Condition C. The proof requires only two lines from the fixed point condition and Corollary 5.

10 Conclusion

We have developed a complete theory of policy improvement under function approximation, centered on the certification gap $\hat{\Delta}_k = J(\mu_k) - \nu^\top f_{k+1}$: the exact critic bias, computable from rollout data.

The Generalized PDL (Theorem 7) establishes the identity $J(\mu_{k+1}) - J(\mu_k) = A - \hat{\Delta}_k$ for any critic and any policy update, making explicit what CRPI bounds (Term A) and what it misses ($\hat{\Delta}_k$). The phase transition (Theorem 20) and efficiency formula (Theorem 22) show that the CRPI bound is binary — either certifying improvement with efficiency $(1 - 1/\tilde{\rho}_k)^2$ above the threshold $\tilde{\rho}_k = 1$, or certifying nothing below it. The relative criterion (Theorem 28) establishes $\epsilon_u \leq c \cdot A_k$ as the correct and necessary scaling condition for maintained certification throughout training, and shows concretely that fixed tolerances cause the policy to freeze below optimal.

The geometric contribution — that FA changes the dominant error from second-order to first-order in step size, requiring TV geometry rather than KL geometry — explains why TRPO's natural gradient is suboptimal under function approximation: it constrains the wrong quantity. The TV-natural gradient $M(\theta)^{-1}g^f$ strictly dominates $F(\theta)^{-1}g^f$ in certified improvement ratio (Theorem 36), with equality only at the uniform policy.

All four contributions are validated to machine precision through experiments that test exact theorems directly: the efficiency formula (Figure 1), the Generalized PDL identity (Figure 3), the TV/KL dominance (Figure 6), and the consequence of the relative criterion for policy convergence (Figure 7).

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A Proof of Theorem 16

Proof. *(i)* Using $f_{t+1} = (1-\alpha)f_t + \alpha T^{\mu_k} f_t$ and linearity of T^{μ_k}:

$$T^{\mu_k} f_{t+1} - f_{t+1} = [(1-\alpha)I + \alpha\gamma P^{\mu_k}](T^{\mu_k} f_t - f_t).$$

The matrix $(1-\alpha)I + \alpha\gamma P^{\mu_k}$ has non-negative entries since $\alpha \leq 1$ gives $1-\alpha \geq 0$ and $P^{\mu_k} \geq 0$. By induction $T^{\mu_k} f_t - f_t \geq 0$, so the product is ≥ 0.

(ii) $\hat{\Delta}_{k,t+1} = J(\mu_k) - \nu^\top f_{t+1} = J(\mu_k) - \nu^\top f_t - \alpha\nu^\top(T^{\mu_k} f_t - f_t) = \hat{\Delta}_{k,t} - \alpha\nu^\top(T^{\mu_k} f_t - f_t)$.

(iii) From (i): $T^{\mu_k} f_t - f_t \geq 0$; since $\nu \geq 0$: $\nu^\top(T^{\mu_k} f_t - f_t) \geq 0$, giving $\hat{\Delta}_{k,t+1} \leq \hat{\Delta}_{k,t}$. In the tabular setting TD converges to $f^* = Q^{\mu_k}$, so $\hat{\Delta}_{k,t} \to 0$. □

B Proof of Proposition 26

Proof. Take $S = 1$, $A = 2$, $r = (0,0)^\top$, $P(s|s,a) = 1$ for all a. Then $Q^\mu = 0$ for every μ, so $J(\mu) = 0$. Let f satisfy the RPI invariant $T^{\mu_k} f \geq f$ with $\epsilon_u = \epsilon^{\text{fixed}}$. Since $T^{\mu_k} f(a) = \gamma \bar{f}^{\mu_k}$ for both a and $\bar{f}^{\mu_k} = \sum_{a'} \mu_k(a') f(a')$, let $f(a_1) > f(a_2)$ so $\bar{\mu}_k = a_1$. Then $A_k = \delta_k(f(a_1) - f(a_2))$, $C_k = 2\delta_k$, giving $A_k/C_k = (f(a_1) - f(a_2))/2$. The RPI condition at a_1 gives $\gamma \bar{f}^{\mu_k} \geq f(a_1)$, from which $f(a_1) - f(a_2) \leq \epsilon_u$. Therefore $\tilde{\rho}_k = 2(1-\gamma)A_k/(C_k\epsilon_u) = (1-\gamma)(f(a_1) - f(a_2))/\epsilon_u \leq 1-\gamma < 1$. □

C Additional Figures

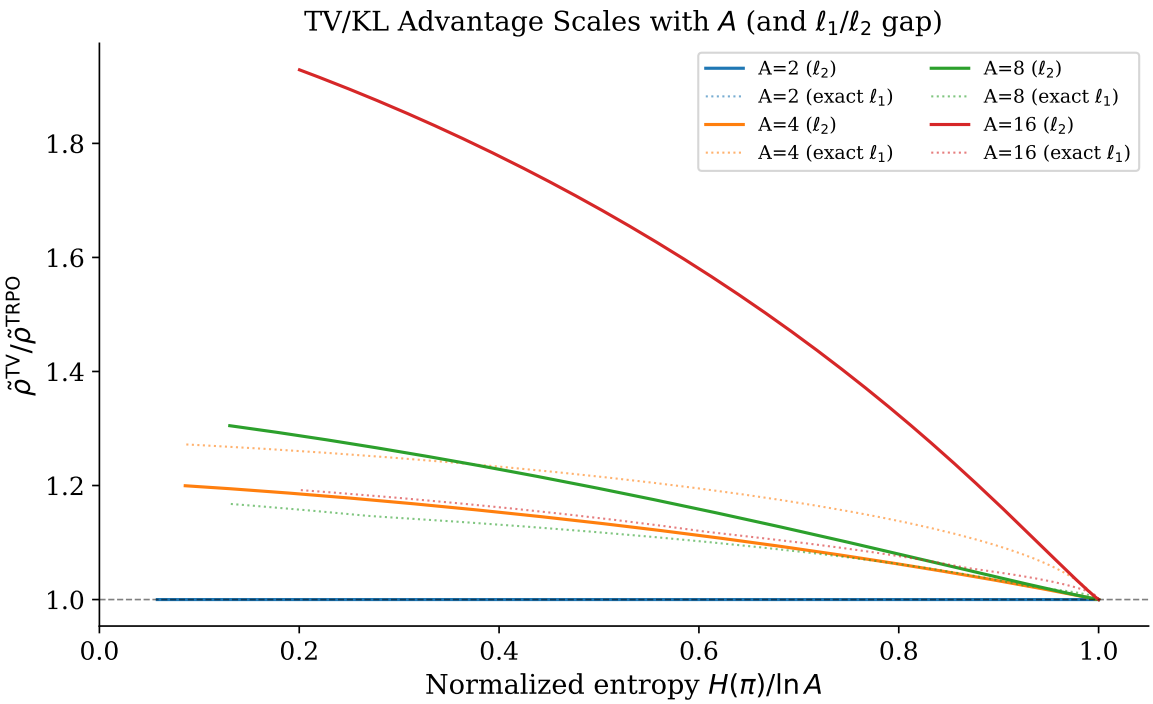

Figure 8: TV/KL advantage scales with A: the ℓ_2 approximation of Theorem 36 is exact at $A = 2$ and introduces an error bounded by $\sqrt{A}$.